LDA vs PCA

1. Which statement correctly distinguishes PCA from LDA?
    - A) PCA is supervised, while LDA is unsupervised
    - B) PCA is unsupervised, while LDA is supervised
    - C) Both are supervised techniques
    - D) Both are unsupervised techniques
2. What information does PCA primarily use to determine the projection directions?
    - A) Class labels
    - B) Class means
    - C) Variance in the data
    - D) Class probabilities
3. What information does LDA use that PCA does not?
    - A) Feature values
    - B) Class labels
    - C) Number of samples
    - D) Number of features
4. The primary objective of PCA is to:
    - A) Maximize class separation
    - B) Minimize within-class variance
    - C) Maximize variance in the projected data
    - D) Maximize classification accuracy
5. The primary objective of LDA is to:
    - A) Maximize total variance
    - B) Maximize between-class separation while minimizing within-class variation
    - C) Minimize the number of samples
    - D) Remove all correlated features
6. Which technique is generally more suitable when the main goal is classification-oriented dimensionality reduction?
    - A) PCA
    - B) LDA

- C) K-Means
- D) Apriori

7. Which technique can be applied when class labels are unavailable?
   - A) LDA only
   - B) PCA only
   - C) Both PCA and LDA
   - D) Neither PCA nor LDA
8. Consider a dataset where the direction of maximum variance does not correspond to the direction that best separates the classes. Which method is more likely to find the class-separating direction?
   - A) PCA
   - B) LDA
   - C) K-Means
   - D) Linear Regression
9. Which matrices are central to LDA but not required in the same way for standard PCA?
   - A) Within-class and between-class scatter matrices
   - B) Adjacency and distance matrices
   - C) Precision and recall matrices
   - D) Confusion and covariance matrices
10. PCA generally finds principal components by performing:
    - A) Eigenvalue decomposition of the covariance matrix
    - B) Classification using class means
    - C) Clustering based on labels
    - D) Regression on target variables
11. LDA commonly obtains discriminant directions using:
    - A) $S_W^{-1}S_B$
    - B) $S_B^{-1}S_W$only
    - C) The identity matrix
    - D) The confusion matrix
12. For a dataset with Cclasses, the maximum number of meaningful LDA components is:
    - A) C
    - B) $C + 1$
    - C) $C - 1$

- D) Equal to the number of features

13. The maximum number of principal components produced by PCA is generally related to:
    - A) Number of classes
    - B) Number of features
    - C) Number of labels
    - D) Number of class means
14. Which method attempts to preserve the directions containing the greatest overall variance?
    - A) LDA
    - B) PCA
    - C) Both always
    - D) Neither
15. Which method explicitly attempts to reduce within-class scatter?
    - A) PCA
    - B) LDA
    - C) Both
    - D) Neither
16. Suppose two classes have very different means but their largest variance occurs in a direction where the classes overlap heavily. Which method is more likely to prioritize class separation?
    - A) PCA
    - B) LDA
    - C) Both equally
    - D) Neither
17. Which statement about PCA and LDA is correct?
    - A) PCA always produces better classification accuracy than LDA
    - B) LDA always produces better classification accuracy than PCA
    - C) Their performance depends on the characteristics and assumptions of the dataset
    - D) PCA and LDA are mathematically identical
18. Which technique is more affected by the absence of class labels?
    - A) PCA
    - B) LDA
    - C) Both equally
    - D) Neither

19. A researcher wants to visualize a labeled dataset in 2D while maximizing separation between different classes. Which method would generally be the more appropriate first choice?
    - A) PCA
    - B) LDA
    - C) K-Means
    - D) Apriori
20. A researcher wants to reduce the dimensionality of an unlabeled dataset while retaining as much overall variance as possible. Which method should be preferred?
    - A) LDA
    - B) PCA
    - C) Logistic Regression
    - D) QDA
21. Which statement is NOT correct?
    - A) PCA does not require class labels
    - B) LDA requires class labels
    - C) PCA directly maximizes class separability
    - D) LDA considers class information
22. If the dataset contains 3 classes, what is the maximum number of LDA projection dimensions?
    - A) 1
    - B) 2
    - C) 3
    - D) Equal to the number of original features
23. Which method is generally more appropriate as a preprocessing technique when the goal is noise reduction without using class labels?
    - A) LDA
    - B) PCA
    - C) QDA
    - D) Logistic Regression
24. Which comparison is correct?
    - A) PCA → maximize variance; LDA → maximize class separability
    - B) PCA → maximize class separability; LDA → maximize variance
    - C) PCA → supervised; LDA → unsupervised
    - D) PCA → classification only; LDA → clustering only

25. If the direction of maximum variance in a dataset is unrelated to class discrimination, what is the main potential disadvantage of using PCA for classification?
    - A) PCA may preserve high-variance features that do not help distinguish classes
    - B) PCA cannot perform dimensionality reduction
    - C) PCA requires class labels
    - D) PCA always removes the class-discriminative information
26. Which technique is more directly designed to find a projection where samples belonging to the same class are close and different classes are far apart?
    - A) PCA
    - B) LDA
    - C) K-Means
    - D) Standardization
27. Which statement best summarizes the difference between PCA and LDA?
    - A) PCA focuses on data variance, whereas LDA focuses on class discrimination
    - B) PCA focuses on class discrimination, whereas LDA focuses on data variance
    - C) Both focus only on reducing computational complexity
    - D) Both require class labels and optimize the same objective
28. In a classification problem, which combination can be useful when the original feature space has a very high number of dimensions?
    - A) PCA followed by LDA
    - B) LDA followed by PCA only
    - C) PCA followed by K-Means only
    - D) PCA followed by data deletion
29. Why might PCA be applied before LDA in a high-dimensional dataset?
    - A) To increase the number of classes
    - B) To reduce dimensionality and help address singularity of the within-class scatter matrix
    - C) To remove class labels
    - D) To convert the problem into clustering
30. Which pair is correctly matched?
    - A) PCA — Unsupervised — Variance Maximization
    - B) PCA — Supervised — Class Separation
    - C) LDA — Unsupervised — Variance Maximization
    - D) LDA — Unsupervised — Clustering